

S. Dinesh

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Professional Summary

Full Stack Developer with hands-on expertise in MERN stack architecture, scalable cloud infrastructure, and real-time backend services. Proven ability to engineer low-latency web applications and deploy fault-tolerant API integrations. Seeking a software engineering role to build high-performance, distributed systems.

Education

Bachelor of Engineering (B.E.) in Artificial Intelligence and Machine Learning | Meenakshi College of Engineering, Chennai, Tamil Nadu | CGPA: 8.15 | Expected Graduation: 2027

Technical Skills

- **Programming Languages:** Python, JavaScript
- **Frontend Frameworks & UI:** React.js, HTML5, CSS3, Tailwind CSS
- **Backend & APIs:** Node.js, Express.js, REST API Development
- **Databases:** MongoDB, PostgreSQL, SQL, Firestore, Supabase
- **Cloud, DevOps & Tools:** AWS, GCP, Azure, Docker, Cloudflare, n8n, Git, GitHub, Vercel, Render
- **AI & Machine Learning:** TensorFlow, Prompt Engineering, LLM APIs

Technical Projects

ChatUp | Real-Time Chat Application

Technologies: React.js, Node.js, Express.js, MongoDB, Socket.io

- Engineered a real-time messaging architecture utilizing Socket.io and Node.js, capable of sustaining 10,000 to 20,000 concurrent user connections per Node instance using optimized in-memory data structures.
- Achieved a reliable server-side message processing latency of 15-40ms by securely synchronizing Socket.io emissions with synchronous MongoDB database insertions and delivery status updates.
- Designed the backend session management architecture with a clear migration path toward horizontal scaling via Redis clustering to overcome single-thread CPU limitations.

Roasting AI | AI-Powered Roast Generator

Technologies: React.js, Node.js, @google/generative-ai SDK, LLMs

- Integrated the Gemini API via the @google/generative-ai SDK, achieving rapid average generation response times of 1.5 to 2 seconds utilizing the gemini-3.1-flash-lite model.
 - Architected a highly resilient backend API featuring a cascading LLM fallback mechanism, implementing strict 8-second timeouts to mitigate primary server load.
 - Engineered fault-tolerant routing logic to seamlessly transition failed requests from primary APIs to token-injected, open-weights fallback models (gemma-4-26b-a4b-it and gemma-4-31b-it).
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